The GSS Mixing Pipeline:
components, parameter choices, and their rationales

local-mixing project

2026-08-05

For readers familiar with the modules who want to understand what the knobs are and why they are set the way they are. Deep dives are referenced by filename (all in `docs/` unless noted). The runnable packaging of everything here is `scripts/gss_mix.sh`; see `GSS_MIX.md`. The Markdown twin of this document is `PIPELINE_OVERVIEW.md`.

$n \to$ **I. construction** $\to$ GSS ($4n$ wires) $\to$ **II. phase A** $\to$ **III. phase B** $\to$ **IV. fcompress** $\to$ deliverable

The pipeline's one free input is n, the wire count of the secret computation C. Everything else is a library convention (a function of n), a validated preset, or a measured calibration — this document says which, and why.

1 Construction: sliced sandwich + gadgetization

Sample a fresh random `g57` circuit C on n wires; build the sliced sandwich A on $2n$ wires (random D, keyed slice interleave, N-column float) with $A(x, 0) = (\text{junk}, C(x))$ on the zero slice; gadgetize to $4n$ wires behind a zero-slice preblock with the diversified CG menu, nonlinear RGs, the product-share mask encoding, and a final commuting shuffle (`SLICED_SANDWICH.md`, `NEW_GADGETIZE.md`, `NONLINEAR_RG_CG_MENU.md`).

knob	value	rationale
$\lvert C\rvert = \lvert D\rvert$	round$(n \log_2^2 n)$	library convention; D sized like C so neither sandwich side is thin
slicing budget s	round$(n \log_2 n)$	enough keyed slice mixing to decorrelate block boundaries
slice gates	$20n$	zero-slice preblock size, tool default
mask plan	$[2,2,2,3]$ + Gray fold	measured: leak is LINEAR in the piling-up product ε ($R^2 = 0.996$); this plan cuts ε $3\times$ vs $[2,3,3]$ at -14% gates; disjoint deg-2 piling-up is proved optimal for sparse-ANF masks (`CORRELATING_TWO_COMPUTATIONS.md`, `TWISE_INDEPENDENT_MASKS.md`, `GRAY_FOLD_CG.md`)
Gray fold	on (default)	fold gates ≤ 2 controls $\Rightarrow$ DB reachability $31.6\% \to 95.6\%$; judge digestibility by reachability, never match rate
carrier decode	single-carrier	the old $[3,3]$ was really $[1,1,3,3]$: a second linear carrier free to an affine adversary (`SINGLE_CARRIER_CONSTRUCTION.md`, `SINGLE_CARRIER_DECODE.md`)
band	rolling, nonlinear fill, retire-refill epochs	band-in-slice exact disturbance theorem; rolling $+12\%$; fill must be jointly uniform; band *lifetime*, not width, is the recovery channel (`BAND_HARDENING.md`, `BAND_INDEPENDENCE.md`)

No `PROD_*` overrides: the production preset IS the default, per `PRODUCT_SHARE_ENCODING.md` and `PRODUCT_SHARE_UPDATE.md`. The gadget sample-verifies low-$2n \equiv A$ on the zero slice at build time.

2 Phase A: the DB re-encoding mix

`fmix --gss --phase-a --profile`: a move walk whose DB moves re-spell windows against the frozen store (MIX grows-if-needed, COMP strictly shrinks) under a layer-2 size profile; `g57`-word twists supply state-frame rotation. Phase A is deliberately *`g57`-preserving*: its job is re-encoding depth and spelling diversity, not structure breaking (`POSTMIX_MANUAL.md` §2, `FMIX_PHASE_A.md`, `FMIX_LAYER1.md`, `FMIX_LAYER2.md`).

knob	value	rationale
size profile	3, 3+H, 3+H+20, R, 1+$\frac{R-1}{2}$ (default $3, 33, 53, 2, 1.5$)	expansion fixed at 3 effs (transport happens during growth — short, steep ramp); hold H=30 at R=2 is the re-encoding workhorse; compression fixed at 20 effs, ending halfway back (1.5×) to keep spelling diversity a full return would spend (`FMIX_LAYER2.md`)
exposed knobs	`--expand`, `--hold`	the only two: peak size and dose
DB knobs	the `--gss` profile	COMP descends from s_{db} 12 (a start: walks 12 . . . 1), MIX draws uniformly ≤ 6; curated-first ON both modes; precedence in `FMIX_PARAM_PRECEDENCE.md`; calibrated by the mode×geometry×curation cube (`DB_CAMPAIGN_20260805.md`, `CURATED_DB_COMPARISON.md`)
twists	`--phase-a`: twist-`g57` at $p = 0.0005$	brackets are adaptive all-`g57` words absorbing neighborhood gates (`G57_TWIST_BRACKETS.md`); rate tiny — one twist rewrites O(window) gates
store guards	degree 9, span 30, terms 1024/2048	the store's max ANF degree; canonicalization cost caps
env	`FROZEN_DB_DIR`, `FROZEN_CURATED_DIR`	frozen-store-only runtime; read at startup, never from rc files

3 Phase B: structure breaking

Phase B's reframed job (2026-08-04) is **anti-inversion**: break the `g57` structure with absolute spread — not state mixing (fmix alone hides nothing; hiding is sandwich-borne). No DB, no swap-family twists.

Part 1 — the split stage (`--split`, shipped defaults). One move to `g57` exhaustion: presplit a random `g57` into (CNOT/NCNOT, 2-control AND); with $p_{\text{join}} = 0.8$ wrap an *absorbed* pure-NOT twist on its target wire — both brackets flip a control polarity (zero synthetic gates), the segment's w-reading pins flip, segment `g57`s force-split — then one cross from the AND piece. Bracket side drawn ∝ remaining circuit length (kills the short-span spike; squared suppression); bracket = farthest of k=2 samples. Growth = 1+ comp-fraction (measured ×1.7–2.0); output is dead-even CNOT/NCNOT + 1:2:1 ANDs, zero comp; coverage canary-verified absolute. Spec `FMIX_SPLIT_TWIST.md`; measurements and the bracket-draw A/B in `SPLIT_TWIST_REPORT.pdf` and `reports/split_trials_20260805/RESULTS.md`.

Part 2 — the crossing walk (resumes the split state). Pure crossings under the thermostat; the 2026-08-05 X-panel set every knob:

knob	value	rationale
damping	$b = 3$, $c = 1$	heavy damping EQUALIZES: median descendants up, tail down — the only knob that lifts the median (judge spread by MEDIAN absolute descendants/span; the Yule tail makes means dishonest); $b > 3$ probed: plateau by $b \approx 3.5$ (single-seed deltas ≤ 0.006); $b = 3$ kept
temperature	target/25	second-order; medium keeps arrival clean
target `--xr`	2 (2.5 = max-spread)	frontier at the arrival peak: realized 1.63/1.84/2.01/2.25× → frac(≥3 desc) 0.49/0.56/0.60/0.62, median span 360/464/552/649 — decelerating; 2 clears median ≥ 3 with margin
budget	6× target; **stop at arrival**	both median metrics PEAK at the damped equilibrium and the constant-size hold ERODES them (frac≥3: 0.555 → 0.498 over +20M moves) — the hold moves spread from the median to the tail; no moves-for-size trade exists past arrival

The 2/3 bar on frac(≥3) is out of expansion's reach; the next lever is the min-dgen cross-shot bias (planned), not more growth.

4 fcompress: the final pass and honesty check

The attacker-computable greedy compressor, run whole-function (`--live-wires all`), as both the final size pass and the honesty metric: the residual is incompressibility *earned* by mixing — healthy mixed material lands $\gtrsim$ 90%, vs 83% for raw split artifacts (`POSTMIX_MANUAL.md` §3). Tool defaults; nothing tuned per run.

5 Quick reference

stage	knob	default	change when...
all	`-n`	(required)	it's the problem size
all	`-s` seed	1	independent replicate wanted
II	`--expand`	2	more peak size for more re-encoding room
II	`--hold`	30 effs	more/less generation dose
III.1	(none exposed)	shipped defaults	—
III.2	`--xr`	2	2.5 for max median spread; decelerates above
III.2	`--xb/--xc/--xtdiv`	3 / 1 / 25	X-panel-calibrated; damping plateaus by $b \approx 3.5$
III.2	`--xmoves`	6× target	never linger — arrival is the peak

Ops: production sizes on the server; store env vars exported in the launch environment; every fmix launch exports `FMIX_STOP_FLAG`/`FMIX_DUMP_FLAG`; outputs promoted to `circuits/` need a `CIRCUIT_GENERATION_INFO` entry.